Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2025

Course: B.Tech *Time: 03 Hours*
Code: 102808 *Subject: : Non-conventional Manufacturing* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):- **[2 x 7 = 14]**

(a) Which of the following is NOT a characteristic of non-conventional machining processes?
(i) High tool wear
(ii) Suitable for hard and brittle materials
(iii) Minimal mechanical contact
(iv) Can machine complex shapes

(b) The main reason for the development of non-conventional machining processes was:
(i) To reduce energy consumption
(ii) To eliminate manual labor
(iii) To machine materials that are difficult to machine by conventional methods
(iv) To improve surface finish only

(c) Which of the following processes is classified under non-conventional machining?
(i) Turning
(ii) Milling
(iii) Electro Discharge Machining
(iv) Drilling

(d) In non-conventional machining, the material removal typically occurs due to:
(i) Shear deformation
(ii) Mechanical cutting
(iii) Physical or chemical energy
(iv) High-speed tool rotation

(e) Which of the following statements about ultrasonic machining (USM) is correct?
(i) USM uses thermal energy to melt the material
(ii) USM is suitable only for soft materials
(iii) USM uses high-frequency vibrations to aid material removal
(iv) USM requires electrical conductivity of the workpiece

(f) In Electro-Discharge Machining (EDM), material is removed primarily by:
(i) Mechanical abrasion
(ii) Chemical dissolution
(iii) Electrical discharge causing localized melting and vaporization
(iv) High-pressure fluid jet

(g) Which of the following is a major requirement for the workpiece material in Electro-Chemical Machining (ECM)?
(i) It must be brittle
(ii) It must be electrically conductive
(iii) It must have high thermal conductivity
(iv) It must be non-metallic

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(h) Which non-conventional welding process uses high-energy photons to fuse materials?

(i) Electron Beam Welding (ii) Laser Beam Welding
(iii) Plasma Arc Welding (iv) Ultrasonic Welding

(i) In explosive welding, the bonding of materials occurs due to:

(i) High-frequency vibrations (ii) Intense localized heat
(iii) A chemical reaction (iv) High-pressure shock wave from an explosion

(j) Which welding process is most suitable for joining materials in deep-sea environments?

(i) Plasma Arc Welding (ii) Underwater Welding
(iii) Electron Beam Welding (iv) Laser Beam Welding

Q.2 (a) Explain the historical development and need for non-conventional machining processes. Include at least two reasons why they became necessary. [7]

(b) Classify non-conventional machining processes and briefly explain the principle of any two types. [7]

Q.3 (a) Explain the working principle of ultrasonic machining (USM) and mention its key applications. [7]

(b) Describe the process of water jet cutting. What are its advantages over traditional cutting methods? [7]

Q.4 (a) Explain the working principle of Electro-Discharge Machining (EDM) and list its main industrial applications. [7]

(b) Describe the process of Electro-Chemical Machining (ECM). How does it differ from EDM in terms of material removal mechanism? [7]

Q.5 (a) Explain the principle and working of laser beam welding. Mention at least two key advantages. [7]

(b) Describe electron beam welding and list its major applications in industry. [7]

Q.6 (a) What is ultrasonic welding? Explain its principle and name one industry where it is commonly used. [7]

(b) Compare plasma arc welding and explosive welding in terms of mechanism, heat source, and applications. [7]

Q.7 (a) Describe the electro-magnetic forming process and explain how it differs from explosive forming. [7]

(b) Define water hammer forming. How is the energy transferred to the workpiece in this method? [7]

Q.8 (a) Explain how non-conventional machining processes (NCMPs) are used in micro-manufacturing applications. Provide examples. [7]

(b) Describe the concept of Micro to Nano finishing. How does this impact the performance of miniaturized components? [7]

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Key fundamentals and advantages of micro-manufacturing
(b) Challenges involved in micro-manufacturing
(c) Working principle of explosive forming
(d) Ultrasonic Welding

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